DO NOW — RECALL

Exam-style data representation problems

Name: _____ Class: _____ Date: _____

1. What does CountAtLeast([5,7,3],5) return?

2. What does O(log n) often describe in binary search?

3. What is the sign bit for a negative 8-bit two's-complement integer?

4. Calculate $0.1000_2 \times 2^3$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2.

2. Model answer

Repeatedly halving the candidate interval.

3. Model answer

1.

4. Model answer

4.

Vocabulary — know these terms

bit width — Number of bits in a representation.

two's complement — Fixed-width signed binary format.

mantissa — Signed fractional significant component.

exponent — Power-of-two scaling value.

precision — Number of significant digits retained.

NEW LEARNING

Read, annotate and explain

Respect the bit width

Write positive 5 using exactly eight bits.

Invert then add one

The result is 11111011_2, representing -5 in 8-bit two's complement.

Interpret floating-point differently

0.1100_2 is fractional 0.75; multiplying by 2^3 gives 6.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Convert −10 to 8-bit two's complement.

Predict

Write 00001010 for +10 and invert to 11110101.

Explain

Add one: 11110110 represents −10.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Convert unsigned binary 00001011 to decimal.

Question 2

Encode −12 in 8-bit two's complement.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Normalise positive $0.0011_2 \times 2^5$ using a two's-complement fractional mantissa.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Eight-bit two's complement can represent decimal +128.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.